Final Report — One Design, Seven AI Coding Agents

Prepared for **Murpheux** · 28 September 2026 · Baseline: the OpenCode v2 configuration (`~/opencode/.config/opencode`)
Deliverables: `~/agent-configs/` (6 tool folders + `_shared/` build system) · per-tool reports `reports/01-06-*.pdf`

1. Executive summary

The cost-aware multi-agent configuration built for OpenCode was re-implemented **natively** for Claude Code, GitHub Copilot CLI, VS Code (Copilot Chat), Cline, OpenAI Codex CLI and Aider. Each tool has its own folder that mirrors its real home-directory layout, with an installer that backs up anything it replaces. Nothing was installed into live locations.

All seven configurations are generated from **one source of truth**:

- the OpenCode `AGENTS.md` (every fact preserved; 277 items checked in each of the six new instruction files);
- eight tool-neutral agent prompts;
- 16 shared workflows;
- 20 portable skills (Agent Skills spec);
- one guardrail rule set with a single guard hook that speaks five hook contracts.

Current config formats were confirmed for each tool (September 2026 versions), and ideas from each awesome list were adopted where they fit. Everything was validated offline:

- parsers for every config format;
- Starlark rules executed with their examples;
- 70 guard self-tests;
- 34 installer smoke checks in throwaway homes;
- the Windows PowerShell hook run for real.

6	35	81	149	70 + 34
new tool configs	new agent definitions	workflows / commands	skill files	tests passing

2. Approach

1. **Research**: six parallel research agents read each tool's official sources on GitHub, because the corporate proxy blocks many AI vendor sites. Each also read its awesome list:
   - awesome-claude-code → Claude Code;
   - awesome-copilot and awesome-copilot.github.com → Copilot CLI and VS Code;
   - awesome-clinerules → Cline;
   - awesome-codex-cli → Codex;
   - Aider community resources (no list was provided).

   The Copilot CLI agent stalled. It was replaced by the installed CLI's own help (v1.0.61) and the `github/docs` references.
2. **Shared layer**: `_shared/build.py` turns the OpenCode baseline into a placeholder template, portable skills, agent bodies and commands. It renders each tool from a small YAML spec and normalises line endings.

3. **Tool layer**: settings, providers, MCP, hooks and permissions were written per tool, using only keys confirmed for the current version.
4. **Verification**: `validate.py` and `install_smoke.sh`, a content-preservation check, and Windows tests.

3. Per-tool overview

Tool	Version	Installs to	Router mechanism	Agents · commands · skills
OpenCode (baseline)	2.0.18	`~/.config/opencode`	primary `router` agent; edit/shell denied	7 · 16 · 20
Claude Code	2.1.283	`~/.claude`	`"agent": "router"` main session on Haiku; hook keeps it read-only	8 · 16 · 20
Copilot CLI	1.0.61	`~/.copilot`	`cop` = `--agent router`; no edit/shell tools	8 · built-ins · 33
VS Code	1.139.1	profile + `~/.copilot`	Router agent: cheaper subagents + premium **handoffs**	8 · 16 prompts · 20
Cline	4.1.21 / CLI 3.0.65	`Documents/Cline` + `~/.cline`	Plan mode + read-only subagents	4 · 16 workflows · 20
Codex CLI	0.157.1	`~/.codex` + `~/.agents/skills`	main thread + custom agents + profile files	7 · 16 skills · 20
Aider	0.86.2	`~/.aider.conf.yml` + `~/.aider`	aliases/modes (`ai-ask` → `ai` → `ai-arch` → `ai-sec`)	— · 4 + 20 loaders · 20

4. How the design maps onto each tool

Capability	OpenCode	Claude Code	Copilot CLI	VS Code	Cline	Codex	Aider
Cheap router	yes	Haiku main agent	gpt-5-mini agent	Medium tier (cost-tier rule)	Plan mode	main on Sol; cheap profile	alias
Router can't mutate	permissions	hook gate	tool list	tool list	Plan mode blocks writes	—	—
Subagents on own model	yes	frontmatter	settings map	model arrays	agents/*.yaml	agents/*.toml	editor/weak model
Pre-tool guard hook	policies	yes	fail-closed	yes	both builds	yes	n/a
Hard deny rules	policies	permissions.deny	managed file	hook	hook / legacy env	exec-policy	options
OS sandbox	—	yes	yes	terminal	—	yes	—
Secret env scrub	policy	ENV_SCRUB	--secret-env-vars	—	op run	env policy	op run
Skills (on demand)	20	20	33	20	20	36	/read-only loaders
MCP (docs + GitHub)	7	5	4 + built-in	5	5	5	none upstream

5. Model tiers

Tier	Claude Code	Copilot CLI	VS Code	Cline	Codex	Aider
cheap (router/explore/review)	haiku	gpt-5-mini · haiku-4.5	GPT-6 Luna · Haiku 4.5	deepseek-flash	gpt-6-luna	deepseek-flash
workhorse (developer/sre)	sonnet · medium	claude-sonnet-5	Claude Sonnet 5	claude-sonnet-5 (Act)	gpt-6-sol · medium	claude-sonnet-5
reasoning (planner/architect)	opus · medium/high	gpt-5.5 · medium/high	GPT-6 Sol / Opus 5.5 (handoff)	gpt-6-sol (Plan)	gpt-6-sol · xhigh	gpt-6-sol (architect)
premium (security, HIGH_RISK only)	opus · high	claude-opus-4.8	Opus 5.5 (handoff)	claude-opus-5.5	gpt-6-astra	claude-opus-5.5

6. Guardrail coverage (shared `guardrails.json`: 33 command rules, 18 protected paths)

Threat	Claude	Copilot CLI	VS Code	Cline	Codex	Aider
Disk wipe / root-home delete	deny + hook	hook (+managed)	hook	hook	forbidden + hook	no auto shell
Force-push / history rewrite	deny + hook	hook + deny-tool	ask + hook	hook	forbidden + hook	manual only
IaC destroy / auto-approve	deny + hook	hook	ask + hook	hook	forbidden	manual only

Secret files (.env, keys, state)	deny + hook + sandbox	hook + sandbox	edit-ask + hook	hook	hook + sandbox	.aiderignore
Secret printing (op read, printenv…)	deny + hook	hook	hook	hook	forbidden	manual only
Remote code piping (curl | sh)	hook	hook	hook	hook	prompt + hook	manual only
Interactive hangs (vim, less, REPL)	hook	hook	hook	hook	hook	n/a
Exfiltration endpoints	sandbox domains	deniedUrls	fetch not auto	—	network off	detect-urls off
Pre-commit secret scan	ggshield in every `commit` workflow. Aider sets `git-commit-verify: true` because its default skips hooks					

7. Coexistence on one machine

- `~/.copilot` is shared by the Copilot CLI and VS Code's Copilot harness. Both folders install **identical** instructions, skills and hooks; agents are target-scoped (`github-copilot` vs `vscode`).
- `~/.agents/skills` (Codex) is also read by the Copilot CLI and Cline `next`. The Copilot and VS Code installers skip identical copies; Codex workflow skills live in Codex-only `~/.codex/skills`.
- VS Code's Copilot harness also reads `~/.claude/agents` and `~/.claude/skills`. `chat.useClaudeMdFile` is off so `CLAUDE.md` isn't loaded twice, but Claude agents will show in VS Code's picker if both are installed.
- Windows + WSL: each harness reads the host it runs on. Run the Windows `install.ps1` (from a local copy, because UNC scripts are blocked by RemoteSigned) and the WSL `install.sh`.

8. Validation summary

Check	Result
Config parsing: 18 JSON/JSONC, 12 TOML, 18 YAML, 207 Markdown frontmatter files	pass
Skill spec (name == folder, kebab-case, description ≤ 1024): 149 files	pass
Codex exec-policy: 37 rules executed; all `match` / `not_match` examples hold	pass
Guard hook self-tests (commands, paths, 5 payload formats, router gate)	70 / 70
Installer smoke tests in throwaway HOMEs (all six tools)	34 / 34
Instruction content preserved vs original AGENTS.md (6 files × 277 items)	0 missing
Windows: real `PreToolUse.ps1` + Python guard (blocks drive-root delete, allows `dotnet test`)	pass
PowerShell parser (6 scripts) · `bash -n` (all shell scripts) · LF line endings enforced	pass
Live runs of each agent CLI	not possible here (Claude/Codex/Aider/Cline not installed; gateway and OpenRouter not reachable from this network)

9. Issues found and fixed during the build

#	Issue	Fix
1	Files authored from Windows had **CRLF** endings (146 files, including the earlier OpenCode config). This breaks shebangs and hooks on macOS/Linux	All normalised to LF; `build.py` now normalises and `validate.py` rejects CRLF
2	`git --no-pager push` / `git -C x push -f` bypassed the push rules	The guard's git regexes now accept global options; tests added
3	The `adr` workflow overwrote the `adr` skill (and would clash with Claude's `/adr`)	Command renamed to `write-adr` in all tools
4	VS Code refuses subagents priced above the caller, so a cheap router can't call premium specialists	Router on the Medium tier; Architect and Security as handoffs

5	Cline subagents bypass approvals	All defined subagents are read-only; mutations stay in Act mode
6	Copilot CLI user settings have no allow/deny rules	Hook + sandbox + aliases; optional root-owned managed policy
7	Codex custom prompts were removed; the 32 KiB instructions cap would truncate AGENTS.md	Workflow skills; `project_doc_max_bytes = 65536`
8	Claude's built-in Explore now inherits the main model; a main-thread router gate would have blocked read-only context commands	Explore overridden to Haiku; read-only allowlist in the router gate
9	Aider commits with `--no-verify` by default (skips ggshield); `~/.env` auto-load would clobber `op run` keys	`git-commit-verify: true`; `~/.aider/aider.op.env`
10	Windows execution policy blocks unsigned scripts on `\\wsl.localhost` paths	Documented: run the installers from a local copy

10. Limitations & assumptions

- **Model ids** follow each tool's catalog at verification time. Confirm them once in each picker; VS Code agents fall back to Auto.
- **Gateway and OpenRouter compatibility**: LiteLLM `/v1/responses` for Codex, Anthropic pass-through for Claude, and `reasoning_effort` passthrough are assumed. They are optional overlays where uncertain.
- **Cline**: provider, Plan/Act model and auto-approve choices live in extension state, so they are documented, not written.
- **Aider prices** in `model.metadata.json` are placeholders.
- **The guard is a strong guardrail, not a sandbox**. The OS sandboxes (Claude, Copilot, VS Code, Codex) and least-privilege credentials remain the real boundary.

11. Action items

#	Action
1	Pick the tools you use and run `DRY_RUN=1 ./install.sh` then `./install.sh` in each folder (Windows: `install.ps1` from a local copy).
2	Point the `op://` references (`aider.op.env`, `cline.op.env`, `mcp.op.env`, Codex/Claude `op read`) at your 1Password vault items.
3	Trust or enable hooks where required: Codex `/hooks`, the Cline "Enable Hooks" toggle, VS Code `chat.useHooks` (set). Claude and Copilot load them automatically.
4	Run `codex --strict-config`, Claude `/agents` and `/hooks`, Copilot `/env`, and VS Code *Chat: Diagnostics* to confirm what loaded.
5	Optional hardening: `github-copilot-cli/install.sh --managed`; WSL2 sandbox packages `bubblewrap socat`.
6	To change a rule or a prompt, edit `_shared/`, then run `build.py` → `validate.py` → `install_smoke.sh`, and re-install.